



4.3 Correlation and linear regression

Statistics and probability · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Correlation measures the strength and direction of a linear relationship between two variables, quantified by Pearson's correlation coefficient r ($-1 \leq r \leq 1$): r close to ± 1 indicates strong linear correlation, r close to 0 indicates weak/no linear correlation. The line of best fit (linear regression line) $y = a + bx$ is found (typically by GDC) using least squares, minimising the sum of squared vertical residuals. It can be used to interpolate within the data range, but extrapolation beyond the data range is unreliable.

Worked example: follow the reasoning

Given the regression line $y = 2.5x + 10$ fitted to data with x from 0 to 20, estimate y when $x = 8$, and comment on reliability.

$$y = 2.5(8) + 10 = 20 + 10 = 30.$$

This is interpolation ($x = 8$ is within the data range 0-20), so the estimate is likely reliable.

Quick check · multiple choice

A correlation coefficient of $r = -0.92$ indicates:

- A A strong positive linear relationship
- B A strong negative linear relationship
- C No linear relationship
- D A weak negative relationship

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 The table below shows hours studied (x) and test score (y) for 6 students: [6]

x : 2, 3, 5, 6, 8, 9

y : 50, 55, 65, 70, 82, 88

(a) State, with reference to the data, whether a linear model seems appropriate.

(b) The regression line is $y = 5.6x + 40.1$ (given). Estimate the score for a student who studies 7 hours.

(c) Explain why it would be unreliable to use this regression line to predict the score for a student who studies 20 hours.

Final answer: _____



Answers and explanation

Question 1 [6]

(a) As x increases, y increases fairly steadily, suggesting a strong positive linear relationship is appropriate.

(b) $y = 5.6(7) + 40.1 = 39.2 + 40.1 = 79.3$.

(c) $x = 20$ is far outside the range of the given data (2 to 9 hours), so this would be extrapolation; the linear relationship may not continue to hold outside the observed range (e.g. scores cannot exceed 100), making the prediction unreliable.

Quick check answer

B. r close to -1 indicates a strong negative (inverse) linear correlation.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.